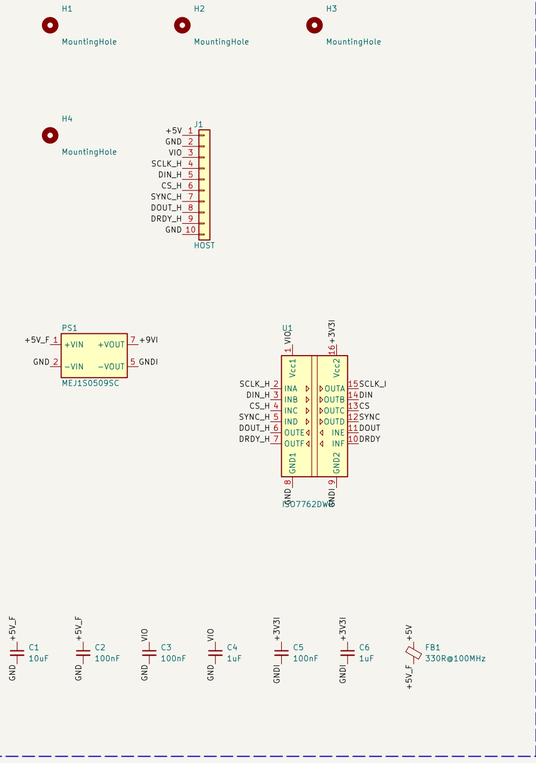
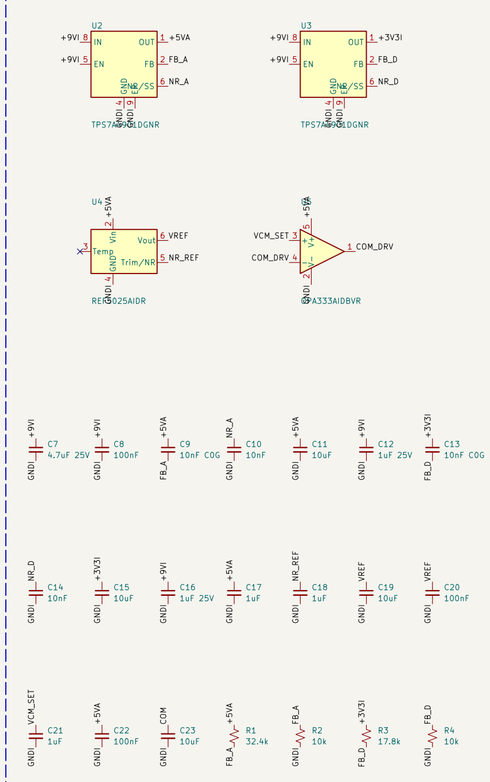
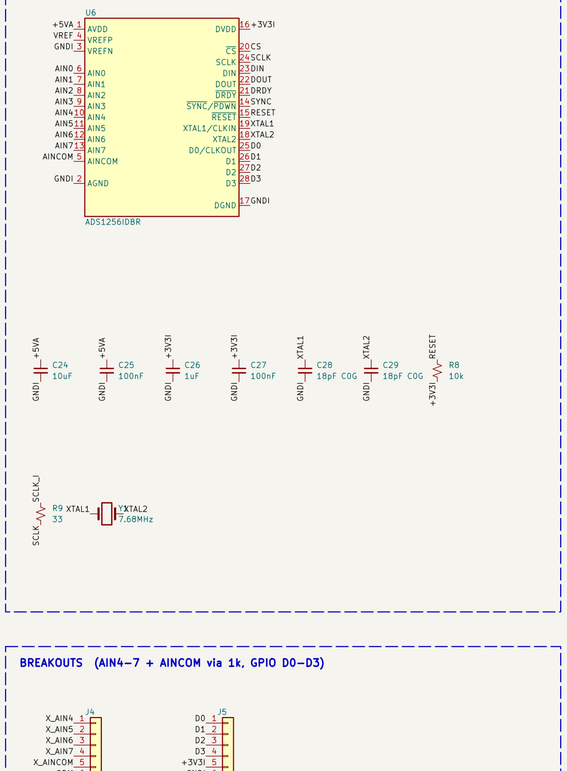
HOST SIDE + ISOLATION BARRIER (ISO7762 4F/2R SPI, MEJ1S0509SC 5 V -> 9 V 5.2 kVDC)



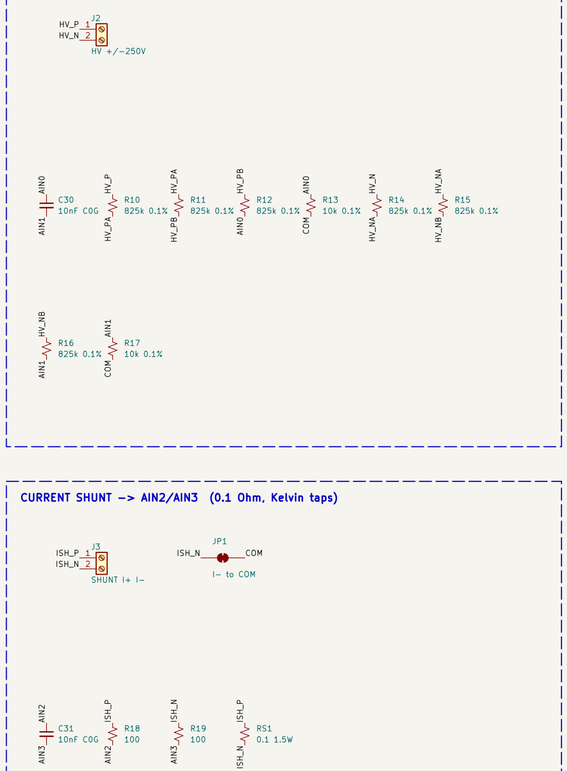
ISOLATED SUPPLIES (AVDD 5.0 V / DVDD 3.3 V TPS7A49, REF5025, COM = 1.5 V buffer)



ADS1256 24-BIT ADC (7.68 MHz crystal)

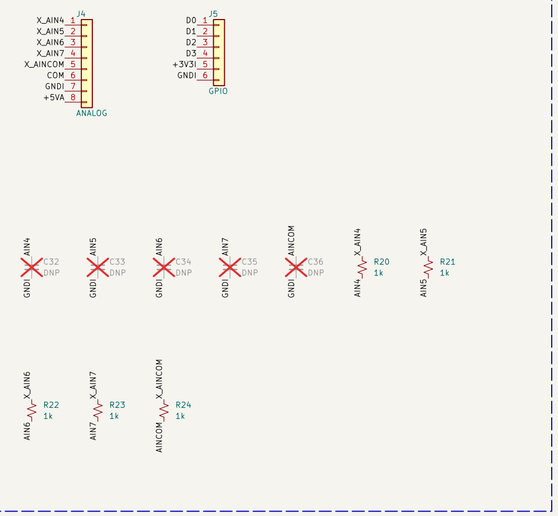


+/-250 VDC INPUT -> AIN0/AIN1 (3 x 825k + 10k per leg, k = 1/248.5, legs return to COM)



CURRENT SHUNT -> AIN2/AIN3 (0.1 Ohm, Kelvin taps)

BREAKOUTS (AIN4-7 + AINCOM via 1k, GPIO D0-D3)



PWR_FLAGs (nets fed from the host header / filters)



Isolated_ADS1256 rev A. Everything right of U1/PS1 is isolated (GNDI). Host side: GND, +5V, VIO.

COM is the isolated-side reference terminal: a buffered 1.5 V above GNDI, centred in the ADS1256 buffered input range (0..3 V).

Connect the measured circuit to COM (JP1 does this at the shunt I- end). Do NOT tie GNDI to the measured circuit too.

AIN0/AIN1: Vin = Vdiff * 248.5. Use BUFEN=1, PGA 4 (+/-1.25 V FS = +/-310 V). Each HV terminal must stay within -+/-370 V of COM.

AIN2/AIN3: I = Vdiff / 0.1 Ohm. PGA 8 = +/-6.25 A FS; RS1 is rated 1.5 W (~3.8 A continuous).

D0/CLKOUT drives 7.68 MHz after reset: clear CLK1:0 in the ADCON register if D0 is unused or used as GPIO.

Generated by tools/gen_sch.py from tools/design.py

OIH Designs

Sheet: /

File: Isolated_ADS1256.kicad_sch

Title: Isolated_ADS1256 - Isolated 24-bit ADC: +/-250 VDC divider, 0.1 Oh

Size: A2	Date: 2026-11-10	**Rev: A**
KiCad E.D.A. 10.99.0-unknown		Id: 1/1